Polynomial Kernel

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Vectors:

$\mathbf{a} = [a_1, a_2]^T$ $\quad\quad$ $\mathbf{b} = [b_1, b_2]^T$

Remember:

$(a^Tb)^2 = a_1^2b_1^2 + 2a_1a_2b_1b_2 + a_2^2b_2^2$

$$(a^Tb + 1)^2 = (a^Tb)^2 + 2(a^Tb) + 1$$

$$= a_1^2b_1^2 + 2a_1a_2b_1b_2 + a_2^2b_2^2 + 2(a_1b_1 + a_2b_2) + 1$$

$$\varphi(a)^T\varphi(b) = [a_1^2, \sqrt{2}\, a_1a_2, a_2^2, \sqrt{2}\, a_1, \sqrt{2}\, a_2, 1] \cdot [b_1^2, \sqrt{2}\, b_1b_2, b_2^2, \sqrt{2}\, b_1, \sqrt{2}\, b_2, 1]$$

Where: $\varphi(x) = [x_1^2, \sqrt{2}\, x_1x_2, x_2^2, \sqrt{2}\, x_1, \sqrt{2}\, x_2, 1]^T$

$\varphi(x) =$

x_1^2

2D ⇒ 6D

$\sqrt{2}\, x_1$

$\sqrt{2}\, x_2$

1